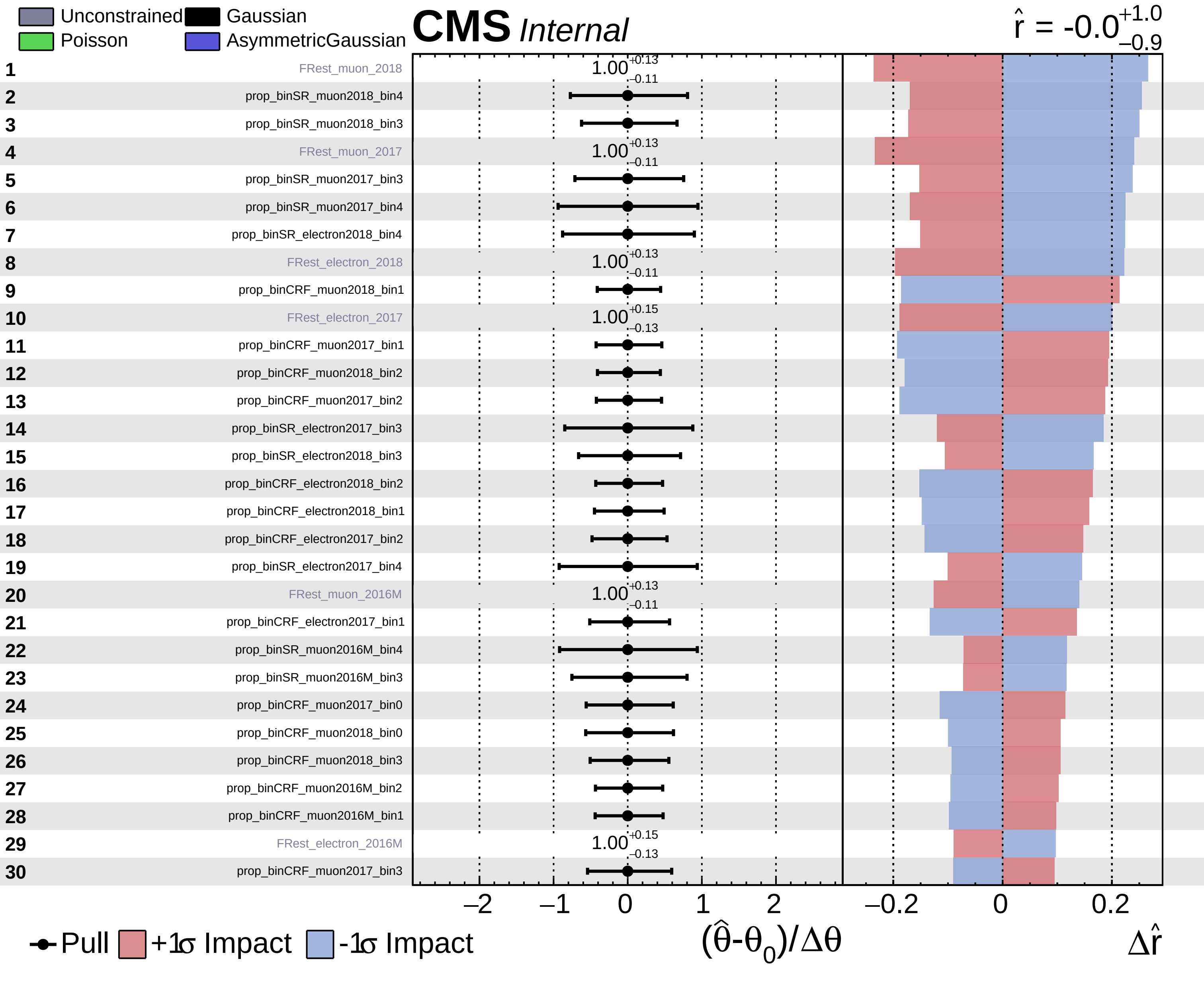
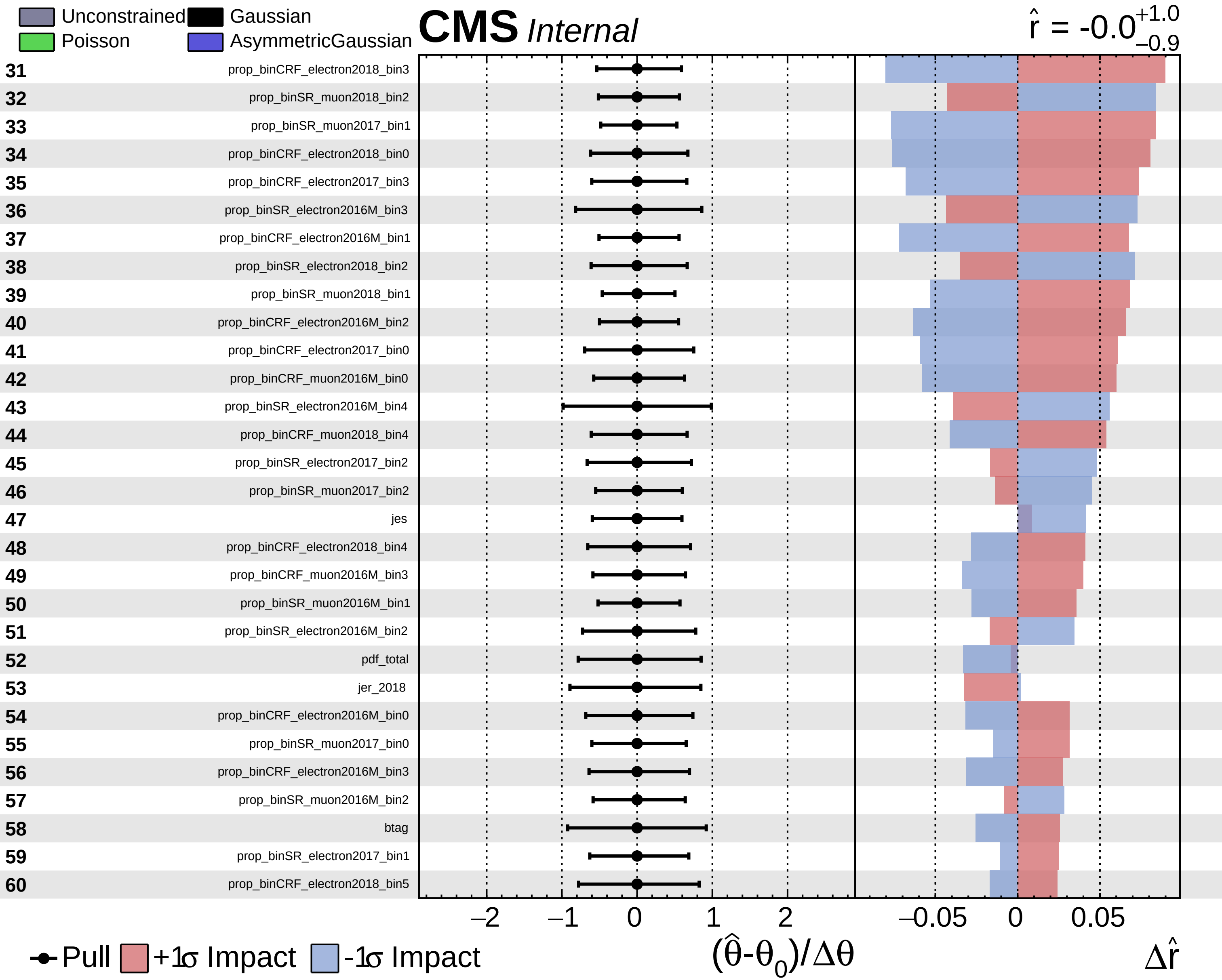
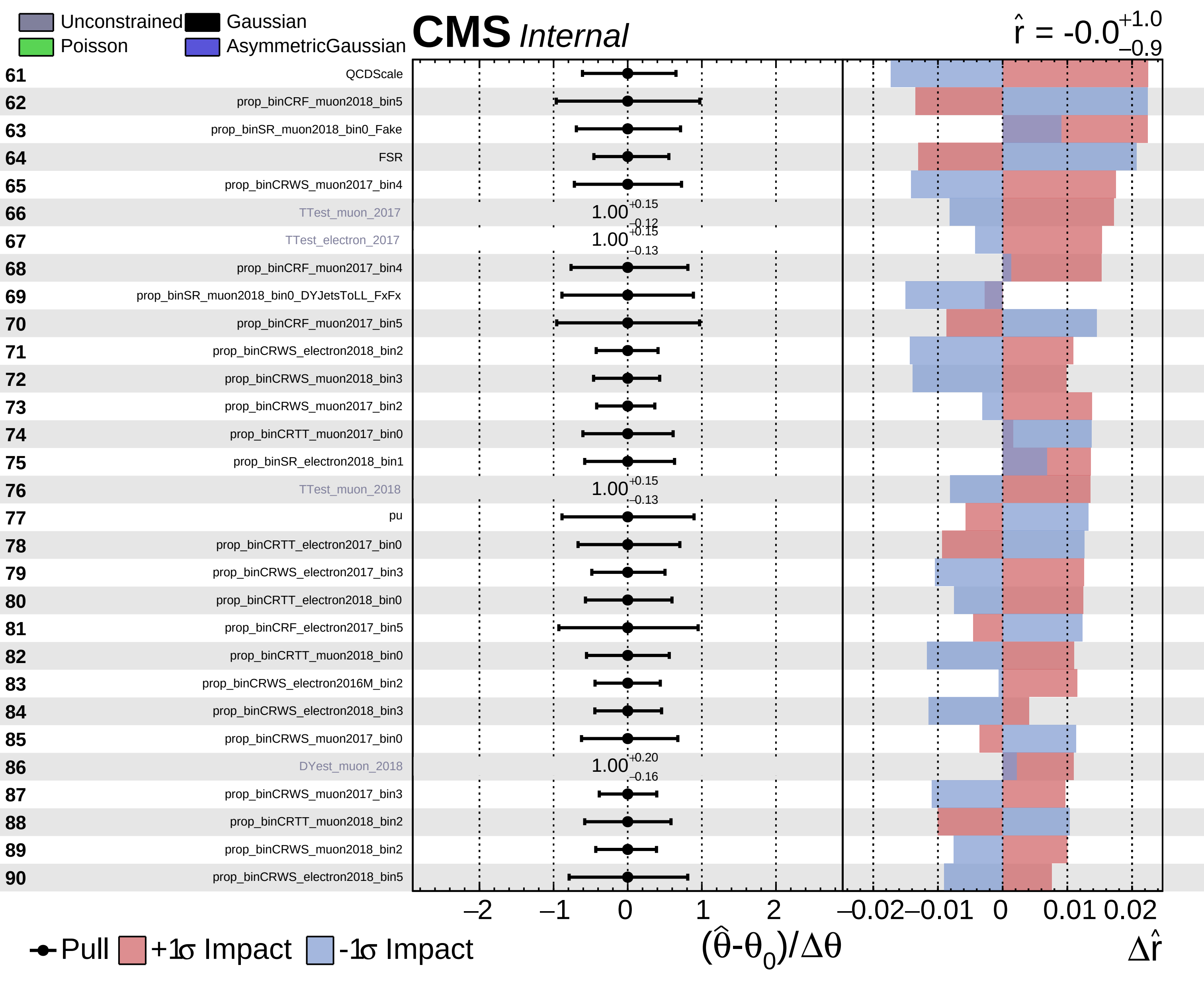


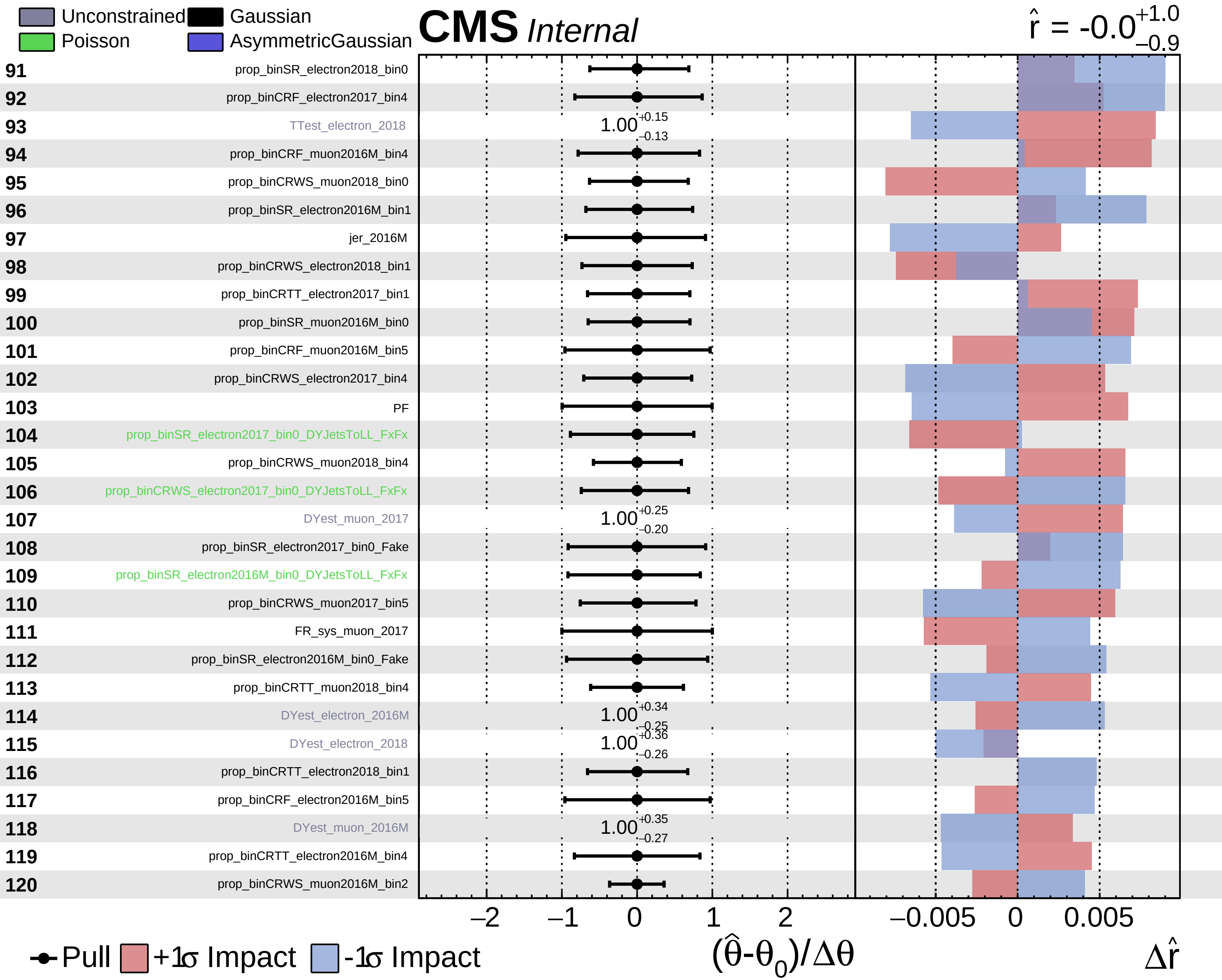
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+1.0}_{-0.9}$
1 FRest_muon_2018 $1.00^{+0.13}_{-0.11}$
2 prop_binSR_muon2018_bin4
3 prop_binSR_muon2018_bin3
4 FRest_muon_2017 $1.00^{+0.13}_{-0.11}$
5 prop_binSR_muon2017_bin3
6 prop_binSR_muon2017_bin4
7 prop_binSR_electron2018_bin4
8 FRest_electron_2018 $1.00^{+0.13}_{-0.11}$
9 prop_binCRF_muon2018_bin1
10 FRest_electron_2017 $1.00^{+0.15}_{-0.13}$
11 prop_binCRF_muon2017_bin1
12 prop_binCRF_muon2018_bin2
13 prop_binCRF_muon2017_bin2
14 prop_binSR_electron2017_bin3
15 prop_binSR_electron2018_bin3
16 prop_binCRF_electron2018_bin2
17 prop_binCRF_electron2018_bin1
18 prop_binCRF_electron2017_bin2
19 prop_binSR_electron2017_bin4
20 FRest_muon_2016M $1.00^{+0.13}_{-0.11}$
21 prop_binCRF_electron2017_bin1
22 prop_binSR_muon2016M_bin4
23 prop_binSR_muon2016M_bin3
24 prop_binCRF_muon2017_bin0
25 prop_binCRF_muon2018_bin0
26 prop_binCRF_muon2018_bin3
27 prop_binCRF_muon2016M_bin2
28 prop_binCRF_muon2016M_bin1
29 FRest_electron_2016M $1.00^{+0.15}_{-0.13}$
30 prop_binCRF_muon2017_bin3
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



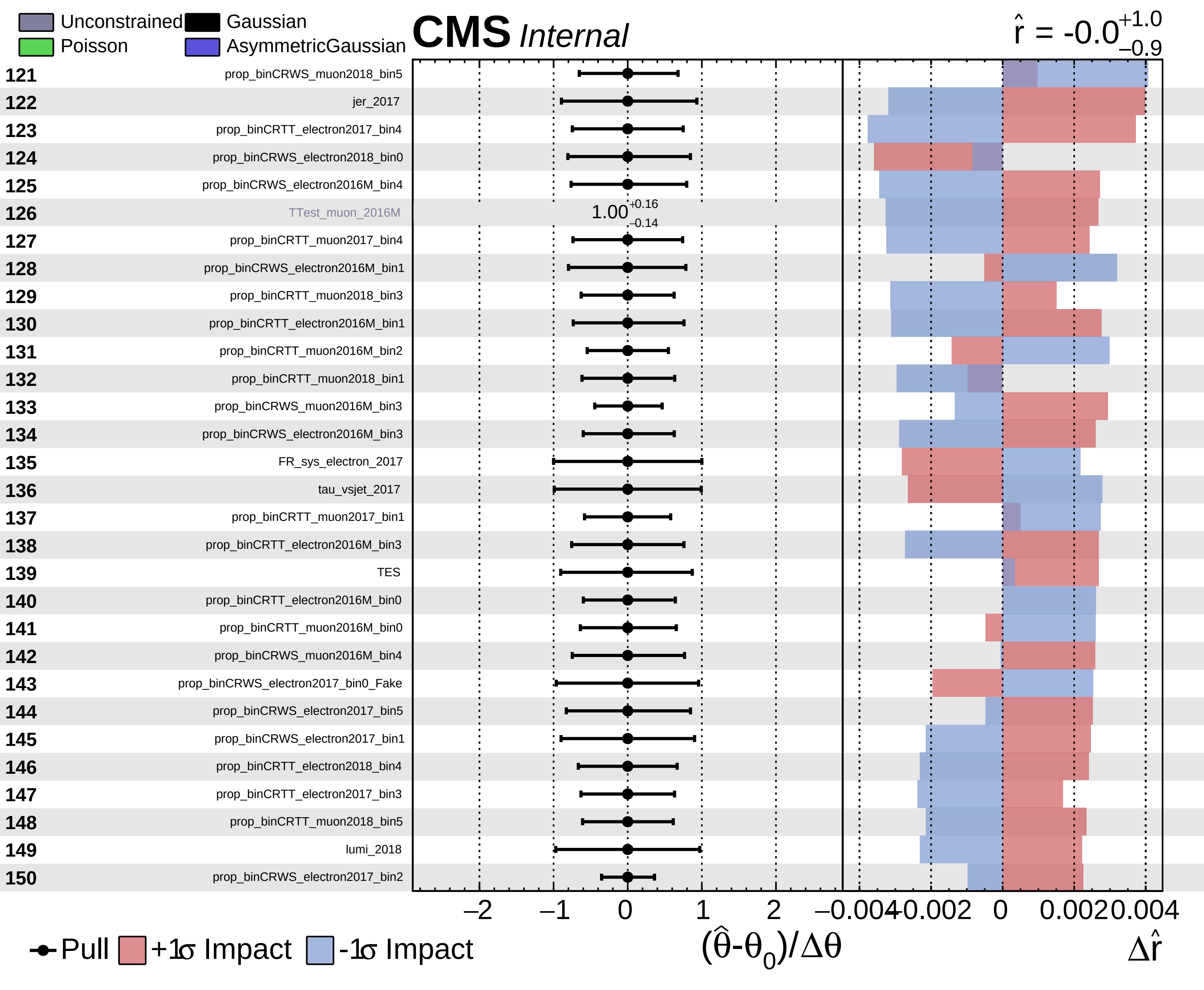
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +1.0 −0.9
31 prop_binCRF_electron2018_bin3
32 prop_binSR_muon2018_bin2
33 prop_binSR_muon2017_bin1
34 prop_binCRF_electron2018_bin0
35 prop_binCRF_electron2017_bin3
36 prop_binSR_electron2016M_bin3
37 prop_binCRF_electron2016M_bin1
38 prop_binSR_electron2018_bin2
39 prop_binSR_muon2018_bin1
40 prop_binCRF_electron2016M_bin2
41 prop_binCRF_electron2017_bin0
42 prop_binCRF_muon2016M_bin0
43 prop_binSR_electron2016M_bin4
44 prop_binCRF_muon2018_bin4
45 prop_binSR_electron2017_bin2
46 prop_binSR_muon2017_bin2
47 jes
48 prop_binCRF_electron2018_bin4
49 prop_binCRF_muon2016M_bin3
50 prop_binSR_muon2016M_bin1
51 prop_binSR_electron2016M_bin2
52 pdf_total
53 jer_2018
54 prop_binCRF_electron2016M_bin0
55 prop_binSR_muon2017_bin0
56 prop_binCRF_electron2016M_bin3
57 prop_binSR_muon2016M_bin2
58 btag
59 prop_binSR_electron2017_bin1
60 prop_binCRF_electron2018_bin5
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



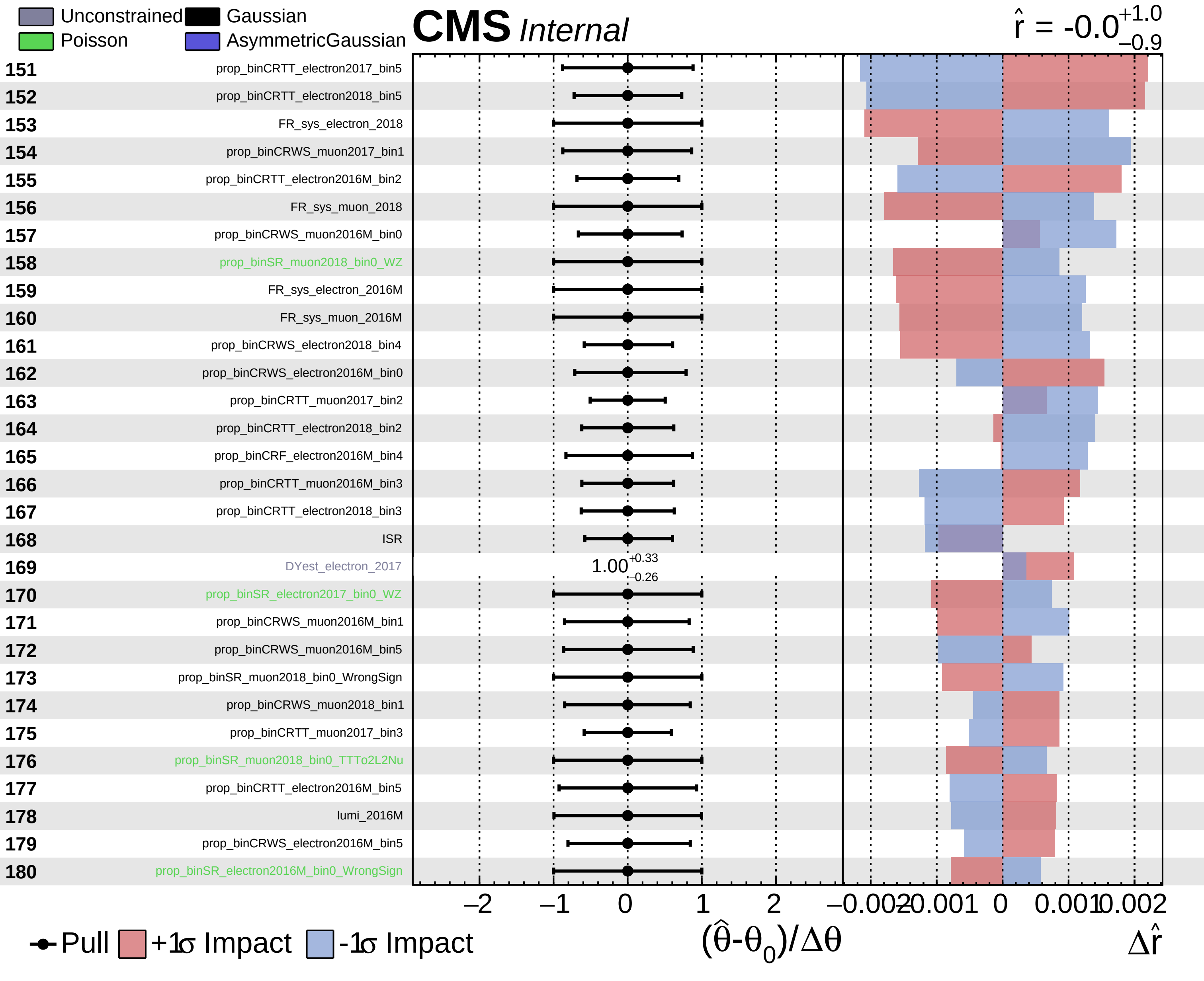
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +1.0 −0.9
61 QCDScale
62 prop_binCRF_muon2018_bin5
63 prop_binSR_muon2018_bin0_Fake
64 FSR
65 prop_binCRWS_muon2017_bin4
66 TTest_muon_2017 1.00 +0.15 −0.12
67 TTest_electron_2017 1.00 +0.15 −0.13
68 prop_binCRF_muon2017_bin4
69 prop_binSR_muon2018_bin0_DYJetsToLL_FxFx
70 prop_binCRF_muon2017_bin5
71 prop_binCRWS_electron2018_bin2
72 prop_binCRWS_muon2018_bin3
73 prop_binCRWS_muon2017_bin2
74 prop_binCRTT_muon2017_bin0
75 prop_binSR_electron2018_bin1
76 TTest_muon_2018 1.00 +0.15 −0.13
77 pu
78 prop_binCRTT_electron2017_bin0
79 prop_binCRWS_electron2017_bin3
80 prop_binCRTT_electron2018_bin0
81 prop_binCRF_electron2017_bin5
82 prop_binCRTT_muon2018_bin0
83 prop_binCRWS_electron2016M_bin2
84 prop_binCRWS_electron2018_bin3
85 prop_binCRWS_muon2017_bin0
86 DYest_muon_2018 1.00 +0.20 −0.16
87 prop_binCRWS_muon2017_bin3
88 prop_binCRTT_muon2018_bin2
89 prop_binCRWS_muon2018_bin2
90 prop_binCRWS_electron2018_bin5
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



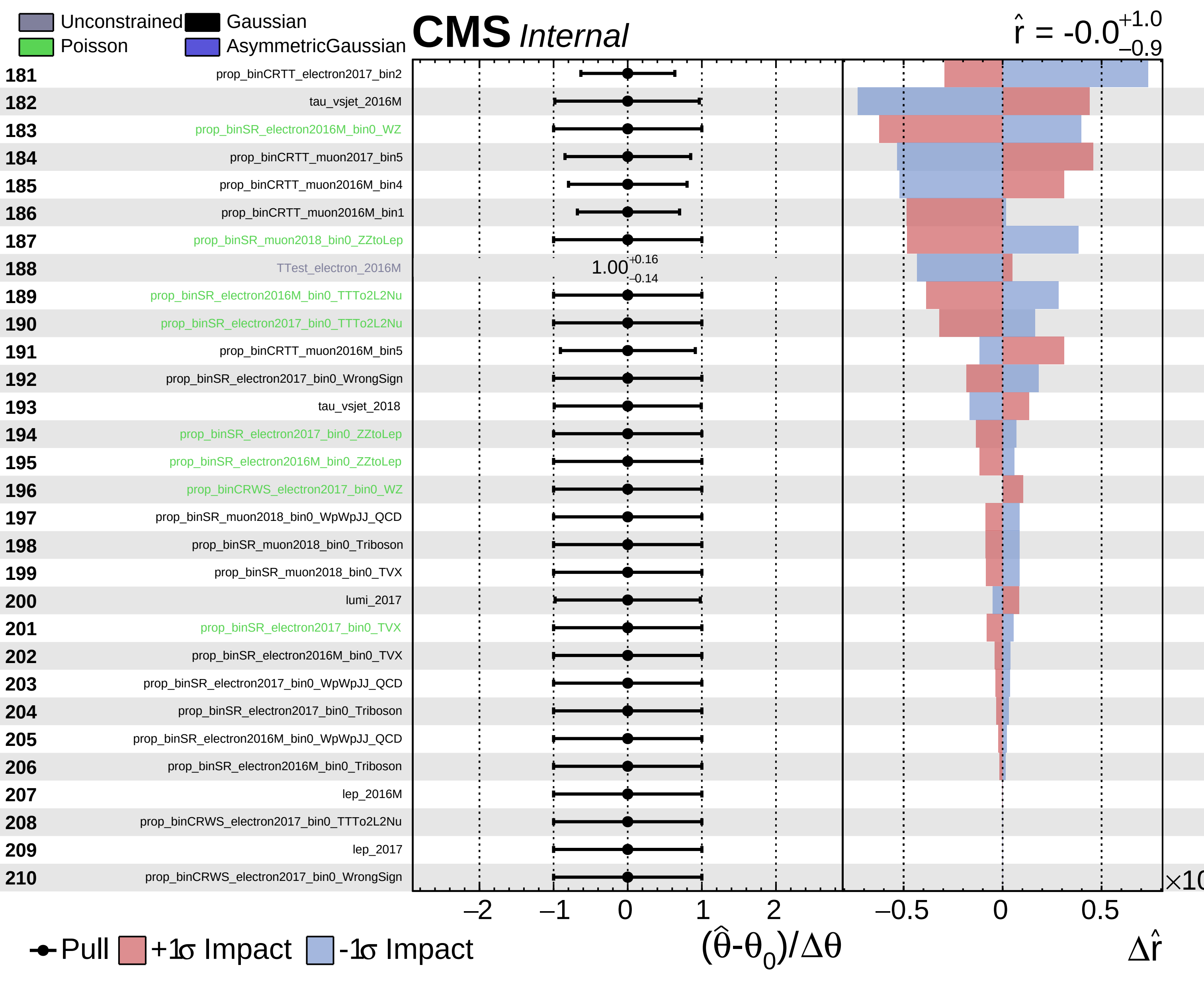
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+1.0}_{-0.9}$
91 prop_binSR_electron2018_bin0
92 prop_binCRF_electron2017_bin4
93 TTest_electron_2018 $1.00^{+0.15}_{-0.13}$
94 prop_binCRF_muon2016M_bin4
95 prop_binCRWS_muon2018_bin0
96 prop_binSR_electron2016M_bin1
97 jer_2016M
98 prop_binCRWS_electron2018_bin1
99 prop_binCRTT_electron2017_bin1
100 prop_binSR_muon2016M_bin0
101 prop_binCRF_muon2016M_bin5
102 prop_binCRWS_electron2017_bin4
103 PF
104 prop_binSR_electron2017_bin0_DYJetsToLL_FxFx
105 prop_binCRWS_muon2018_bin4
106 prop_binCRWS_electron2017_bin0_DYJetsToLL_FxFx
107 DYest_muon_2017 $1.00^{+0.25}_{-0.20}$
108 prop_binSR_electron2017_bin0_Fake
109 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
110 prop_binCRWS_muon2017_bin5
111 FR_sys_muon_2017
112 prop_binSR_electron2016M_bin0_Fake
113 prop_binCRTT_muon2018_bin4
114 DYest_electron_2016M $1.00^{+0.34}_{-0.25}$
115 DYest_electron_2018 $1.00^{+0.36}_{-0.26}$
116 prop_binCRTT_electron2018_bin1
117 prop_binCRF_electron2016M_bin5
118 DYest_muon_2016M $1.00^{+0.35}_{-0.27}$
119 prop_binCRTT_electron2016M_bin4
120 prop_binCRWS_muon2016M_bin2
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



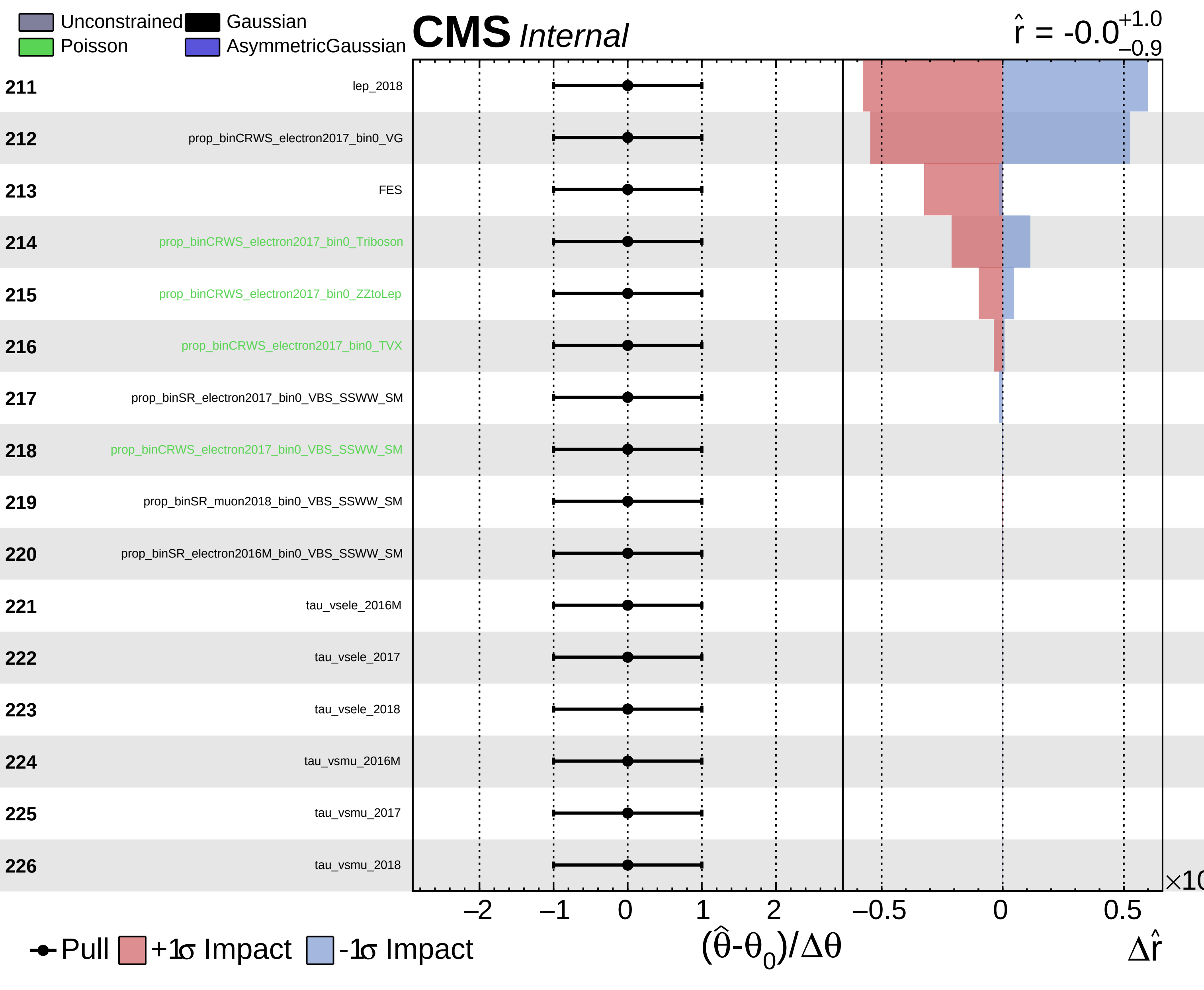
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +1.0 −0.9
121 prop_binCRWS_muon2018_bin5
122 jer_2017
123 prop_binCRTT_electron2017_bin4
124 prop_binCRWS_electron2018_bin0
125 prop_binCRWS_electron2016M_bin4
126 TTest_muon_2016M
1.00 +0.16 −0.14
127 prop_binCRTT_muon2017_bin4
128 prop_binCRWS_electron2016M_bin1
129 prop_binCRTT_muon2018_bin3
130 prop_binCRTT_electron2016M_bin1
131 prop_binCRTT_muon2016M_bin2
132 prop_binCRTT_muon2018_bin1
133 prop_binCRWS_muon2016M_bin3
134 prop_binCRWS_electron2016M_bin3
135 FR_sys_electron_2017
136 tau_vsjet_2017
137 prop_binCRTT_muon2017_bin1
138 prop_binCRTT_electron2016M_bin3
139 TES
140 prop_binCRTT_electron2016M_bin0
141 prop_binCRTT_muon2016M_bin0
142 prop_binCRWS_muon2016M_bin4
143 prop_binCRWS_electron2017_bin0_Fake
144 prop_binCRWS_electron2017_bin5
145 prop_binCRWS_electron2017_bin1
146 prop_binCRTT_electron2018_bin4
147 prop_binCRTT_electron2017_bin3
148 prop_binCRTT_muon2018_bin5
149 lumi_2018
150 prop_binCRWS_electron2017_bin2
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+1.0}_{-0.9}$
151 prop_binCRTT_electron2017_bin5
152 prop_binCRTT_electron2018_bin5
153 FR_sys_electron_2018
154 prop_binCRWS_muon2017_bin1
155 prop_binCRTT_electron2016M_bin2
156 FR_sys_muon_2018
157 prop_binCRWS_muon2016M_bin0
158 prop_binSR_muon2018_bin0_WZ
159 FR_sys_electron_2016M
160 FR_sys_muon_2016M
161 prop_binCRWS_electron2018_bin4
162 prop_binCRWS_electron2016M_bin0
163 prop_binCRTT_muon2017_bin2
164 prop_binCRTT_electron2018_bin2
165 prop_binCRF_electron2016M_bin4
166 prop_binCRTT_muon2016M_bin3
167 prop_binCRTT_electron2018_bin3
168 ISR
169 DYest_electron_2017
$1.00^{+0.33}_{-0.26}$
170 prop_binSR_electron2017_bin0_WZ
171 prop_binCRWS_muon2016M_bin1
172 prop_binCRWS_muon2016M_bin5
173 prop_binSR_muon2018_bin0_WrongSign
174 prop_binCRWS_muon2018_bin1
175 prop_binCRTT_muon2017_bin3
176 prop_binSR_muon2018_bin0_TTTo2L2Nu
177 prop_binCRTT_electron2016M_bin5
178 lumi_2016M
179 prop_binCRWS_electron2016M_bin5
180 prop_binSR_electron2016M_bin0_WrongSign
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +1.0 −0.9
181 prop_binCRTT_electron2017_bin2
182 tau_vsjet_2016M
183 prop_binSR_electron2016M_bin0_WZ
184 prop_binCRTT_muon2017_bin5
185 prop_binCRTT_muon2016M_bin4
186 prop_binCRTT_muon2016M_bin1
187 prop_binSR_muon2018_bin0_ZZtoLep
188 TTest_electron_2016M
1.00 +0.16 −0.14
189 prop_binSR_electron2016M_bin0_TTTo2L2Nu
190 prop_binSR_electron2017_bin0_TTTo2L2Nu
191 prop_binCRTT_muon2016M_bin5
192 prop_binSR_electron2017_bin0_WrongSign
193 tau_vsjet_2018
194 prop_binSR_electron2017_bin0_ZZtoLep
195 prop_binSR_electron2016M_bin0_ZZtoLep
196 prop_binCRWS_electron2017_bin0_WZ
197 prop_binSR_muon2018_bin0_WpWpJJ_QCD
198 prop_binSR_muon2018_bin0_Triboson
199 prop_binSR_muon2018_bin0_TVX
200 lumi_2017
201 prop_binSR_electron2017_bin0_TVX
202 prop_binSR_electron2016M_bin0_TVX
203 prop_binSR_electron2017_bin0_WpWpJJ_QCD
204 prop_binSR_electron2017_bin0_Triboson
205 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
206 prop_binSR_electron2016M_bin0_Triboson
207 lep_2016M
208 prop_binCRWS_electron2017_bin0_TTTo2L2Nu
209 lep_2017
210 prop_binCRWS_electron2017_bin0_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +1.0 −0.9
211 lep_2018
212 prop_binCRWS_electron2017_bin0_VG
213 FES
214 prop_binCRWS_electron2017_bin0_Triboson
215 prop_binCRWS_electron2017_bin0_ZZtoLep
216 prop_binCRWS_electron2017_bin0_TVX
217 prop_binSR_electron2017_bin0_VBS_SSWW_SM
218 prop_binCRWS_electron2017_bin0_VBS_SSWW_SM
219 prop_binSR_muon2018_bin0_VBS_SSWW_SM
220 prop_binSR_electron2016M_bin0_VBS_SSWW_SM
221 tau_vsele_2016M
222 tau_vsele_2017
223 tau_vsele_2018
224 tau_vsmu_2016M
225 tau_vsmu_2017
226 tau_vsmu_2018
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$